MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

RECEIVER, DIFFERENTIAL LINE, DIGITAL, KEY CODE KDN

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

* 1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the procurement of an electronic differential line receiver module.

1.2 Classification. This specification covers two classes of modules;

class I and class II as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.
MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirments . Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.3 Failure rate requirements. All modules procured to this specification shall have a predicted failure rate no greater than 1.4 failures per million operating hours for class I or 2.769 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389. When a class II module is used in a class I environment, failure rate for class I shall be used for reliability prediction (see 6.3).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Electronic Systems Command, ATTN: ELEX 5043, Department of the Navy, Washington, DC 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I. Class I modules shall conform to the electrical requirements of class I only, and class II modules shall conform to the electrical requirements of class I and class II (see 6.3).

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be KDN.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirement for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/53-1 for class I and M28787/53-2 for class II, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance test. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{CC}) of 5.0 ±0.5 Vdc, except that the limits for output short circuit current apply only for a power supply voltage of 5.5 Vdc ±1 percent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

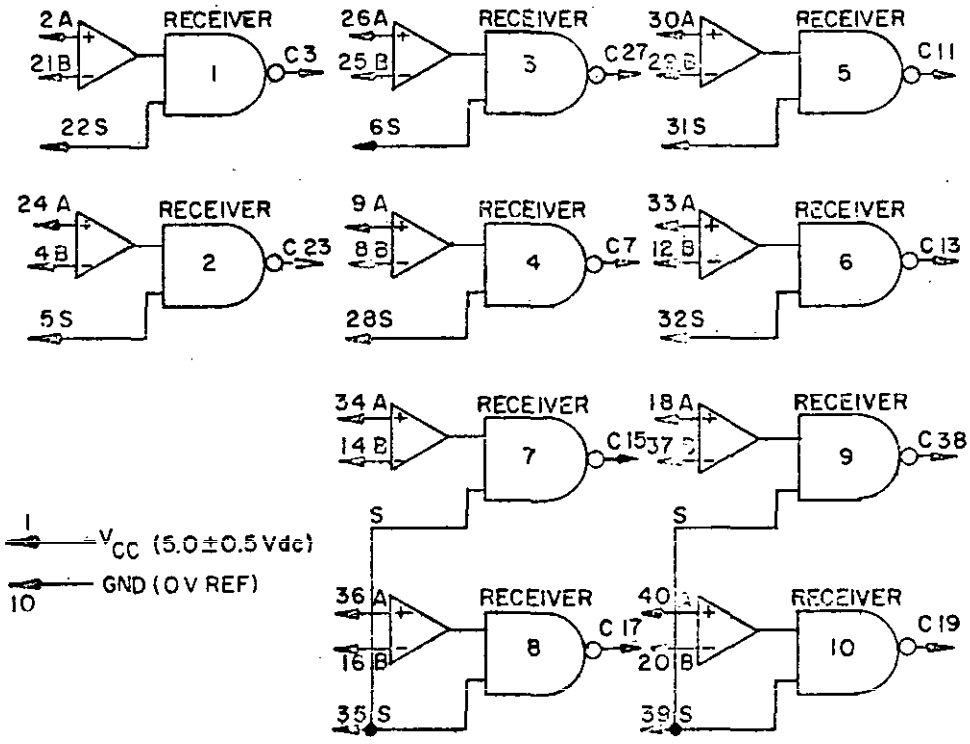
4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I .

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity (SEMP-QAA) and they conform to the requirements of MIL-M-28787.

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* FIGURE 1. KDN module function diagram, differential line receiver.

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* TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	25 +5°C 4/		Class I 4/				Class II 4/				Unit
						0° +5°C		60° +5°C		-55° +5°C		100° +5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
A1-A10	V_{OUT} (low-loaded)	III	S	0.01	0.8	0.01	0.8	0.01	0.8	0.01	0.8	0.01	0.8	V
B	Isolation													
	Initial	IV	S	10	-	10	-	10	-	10	-	10	-	MΩ
	End-of-life	IV	S	1.0	-	1.0	-	1.0	-	1.0	-	1.0	-	MΩ
C1-C29	Input capacitance	V	Q	2	20	2	20	2	20	2	20	2	20	pF
D1-D50	Static output voltage, low level	VI	X	0.0	0.4	0.0	0.4	0.0	0.4	0.0	0.4	0.0	0.4	V
E1-E60	Static output voltage, high level	VII	X	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	V
F1-F20	Input current, +15 V (differential inputs)	VIII	X	2.0	6.0	1.9	6.5	1.9	6.5	1.9	6.5	1.9	6.5	mA
G1-G20	Input current -15 V (differential inputs)	IX	X	2.6	7.8	2.4	8.3	2.4	8.3	2.4	8.3	2.4	8.3	mA
H1-H20	Input current, low level (differential inputs)	X	X	0.1	0.9	0.1	1.0	0.1	1.0	0.1	1.0	0.1	1.0	mA
J1-J6	Input current, low level (strobe)	XI	X	0.5	2.6	0.5	2.6	0.5	2.6	0.5	2.6	0.5	2.6	mA
J7, J8	Input current, low level (common strobe)	XI	X	1.0	5.2	1.0	5.2	1.0	5.2	1.0	5.2	1.0	5.2	mA
K1-K6	Input current, high level (strobe)	XII	X	0.0	2.0	0.0	5.0	0.0	5.0	0.0	5.0	0.0	5.0	µA
K7, K8	Input current, high level (common strobe)	XII	X	0.0	4.0	0.0	10.0	0.0	10.0	0.0	10.0	0.0	10.0	µA
L1-L10	Output short circuit current	XIII	X	15	80	15	80	15	80	15	80	15	80	mA
M1	Power supply current, low level	XIV	S	1	250	1	250	1	250	1	250	1	250	mA
M2	Power supply current, high level	XIV	S	1	250	1	250	1	250	1	250	1	250	mA
N1-N20	Transition time (t_{TLH})	XV	S	4	35	4	40	4	40	4	40	4	40	ns
P1-P20	Transition time (t_{THL})	XV	S	4	15	4	20	4	20	4	20	4	20	ns
R1,R3,R5, R7,R9,R11, R13,R15, R17,R19	Propagation delay time (t_{PHL}) (A inputs)	XV	S	10	53	10	55	10	55	10	55	10	55	ns
R2,R4,R6, R8,R10,R12, R14,R16, R18,R20	Propagation delay time (t_{PLH}) (B inputs)	XV	S	10	57	10	63	10	63	10	63	10	63	ns
S1-S10	Propagation delay time (t_{PHL}) (S inputs)	XVI	S	5	29	5	31	5	31	5	31	5	31	ns

See footnotes at end of table.

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* TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25° ±5°C 4/		Class I 4/				Class II 4/				Unit
						0° ±5°C		60° ±5°C		-55° ±5°C		100° ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
T1,T3,T5, T7,T9,T11, T13,T15, T17,T19,	Propagation delay time (t_{PLH}) (A inputs)	XV	S	10	57	10	63	10	63	10	63	10	63	ns
T2,T4,T6, T8,T10, T12,T14, T16,T18,T20	Propagation delay time (t_{PHL}) (B inputs)	XV	S	10	53	10	55	10	55	10	55	10	55	ns
U1-U10	Propagation delay time (t_{PLH}) (S inputs)	XVI	S	5	26	5	32	5	32	5	32	5	32	ns
V1-V10	Output crosstalk, low level	XVII	Q	0	500	0	500	0	500	0	500	0	500	mV
W1-W10	Output crosstalk, high level	XVIII	Q	0	400	0	400	0	400	0	400	0	400	mV
X	Power supply transient amplitude	XIX	Q	0	500	0	500	0	500	0	500	0	500	mV
Y	Power supply transient rise time	XX	S	1.0	2.0	1.0	2.0	1.0	2.0	1.0	2.0	1.0	2.0	ms
Z1-Z10	Input pulse immunity, high level, check	XXI	Q	0.0	0.4	0.0	0.4	0.0	0.4	0.0	0.4	0.0	0.4	V
AA1-AA10	Input pulse immunity, low level, check	XXI	Q	0.0	0.4	0.0	0.4	0.0	0.4	0.0	0.4	0.0	0.4	V

1/ "X" parameters shall be 100 percent tested during module acceptance test.
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3/ "Q" parameters shall be tested during qualification testing only.
4/ Unless otherwise specified, limits are for initial and end-of-life.

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4.4.2 Order of testing. Test conditions A through AA shall be performed in any order with the following exceptions. If test conditions A, B, or L are part of a sequence of tests, the order of testing shall be as follows: A, B, L where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

* 4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through AA in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions D through AA shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 3.

4.5.1.1 Transient thermal test. The transient thermal test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 4.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 5. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column, and test condition Y, shall be performed using the initial limits at 25°±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3. Following the life test and while still at the maximum class temperature, electrical tests D through AA shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests B through AA shall be performed in the order specified in 4.4.2 using the 25°±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

* 6.3 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in a class I environment, the failure rate for class I shall be used for reliability prediction (see 3.3 and 3.4).

* 6.4 Module description. The following electrical parameters reflect the 0° to 60°C and -55° to 100°C end-of-life limits from table I.

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	Class I	Class II	Unit
Maximum power dissipation (outputs not loaded)	1.38	1.38	W
Input threshold voltage			
Strobe inputs			
Low level	0.8 maximum	0.8 maximum	Vdc
High level	2.0 minimum	2.0 minimum	Vdc
Differential inputs (A and B)			
For 0-7 V common mode applications: When the voltage applied to the A input (V_A) exceeds the voltage applied to the B input (V_B) by +750 mV, or greater, the voltage at A will be recognized as high level and the voltage at B will be recognized as low level.	For 7-15 V common mode applications: When the voltage applied to the A input (V_A) exceeds the voltage applied to the B input (V_B) by +1.0 V, or greater, the voltage at A will be recognized as high level and the voltage at B will be recognized as low level.		
When the voltage applied to the B input (V_B) exceeds the voltage applied to the A input (V_A) by +750 mV, or greater, the voltage at A will be recognized as low level and the voltage at B will be recognized as high level.	When the voltage applied to the B input (V_B) exceeds the voltage applied to the A input (V_A) by +1.0 V, or greater, the voltage at A will be recognized as low level and the voltage at B will be recognized as high level.		
Input current			
Strobe inputs			
Low level (V_{IL} = 0 Vdc)			
Independant strobe inputs	-2.6 maximum	-2.6 maximum	mA
Common strobe inputs	-5.2 maximum	-5.2 maximum	mA
High level (V_{IH} = +4.5 Vdc)			
Independent strobe inputs	5 maximum	5 maximum	µA
Common strobe inputs	10 maximum	10 maximum	µA
Differential inputs (A and B)			
The current flowing out of A or B input, when 0 Vdc is applied to that input and 5.5 Vdc at the other input.	1.0 maximum	1.0 maximum	mA
The current flowing into A or B input, when +15 Vdc is applied to that input and 0 Vdc at the other input.	6.5 maximum	6.5 maximum	mA
The current flowing out of A or B input, when -15 Vdc is applied to that input and 0 Vdc at the other input.	8.3 maximum	8.3 maximum	mA
Output drive			
Low level (V_{OL} = 0.4 Vdc)	15.0 maximum	15.0 maximum	mA
High level (V_{OH} = 2.4 Vdc)	5.0 maximum	5.0 maximum	mA
Input voltage			
Common mode applied to differential inputs (A and B)	15 maximum	15 maximum	Vdc

6.5 Module function. This module contains 10 differential line receivers. This module is intended for use with the differential line driver module, key code KDM (MIL-M-28787/55), in double-ended digital interface applications. Only one KDM circuit may be driven by each KDM circuit. Each circuit has differential inputs designed to reject plus or minus 15 V common mode signals while responding to 750 mV differential signals. Six of the receiver circuits have independent strobes. Two pairs of the remaining four receivers have common strobes. The strobe is active when high.

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6.5.1 Input to output relationship. Each differential line receiver has the input to output relationship of figure 2 and table II for positive logic.

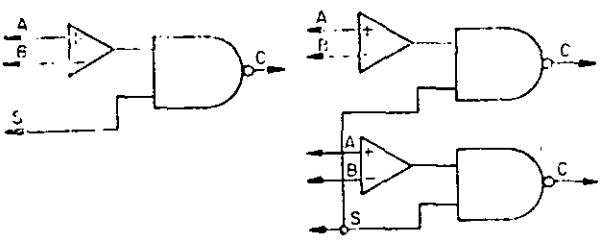


FIGURE 2. Logic diagram of functional circuits.

TABLE II. Input to output relationship.

A	B	S	C
X	X	L	H
L	H	H	H
H	L	H	L
H	H	H	ND
L	L	H	ND

NOTES:
1. X = Don't care condition.
2. L = Low-voltage level.
3. H = High-voltage level.
4. ND = Indeterminate.

6.5.2 Parallel outputs. Paralleling of outputs is not permissible.

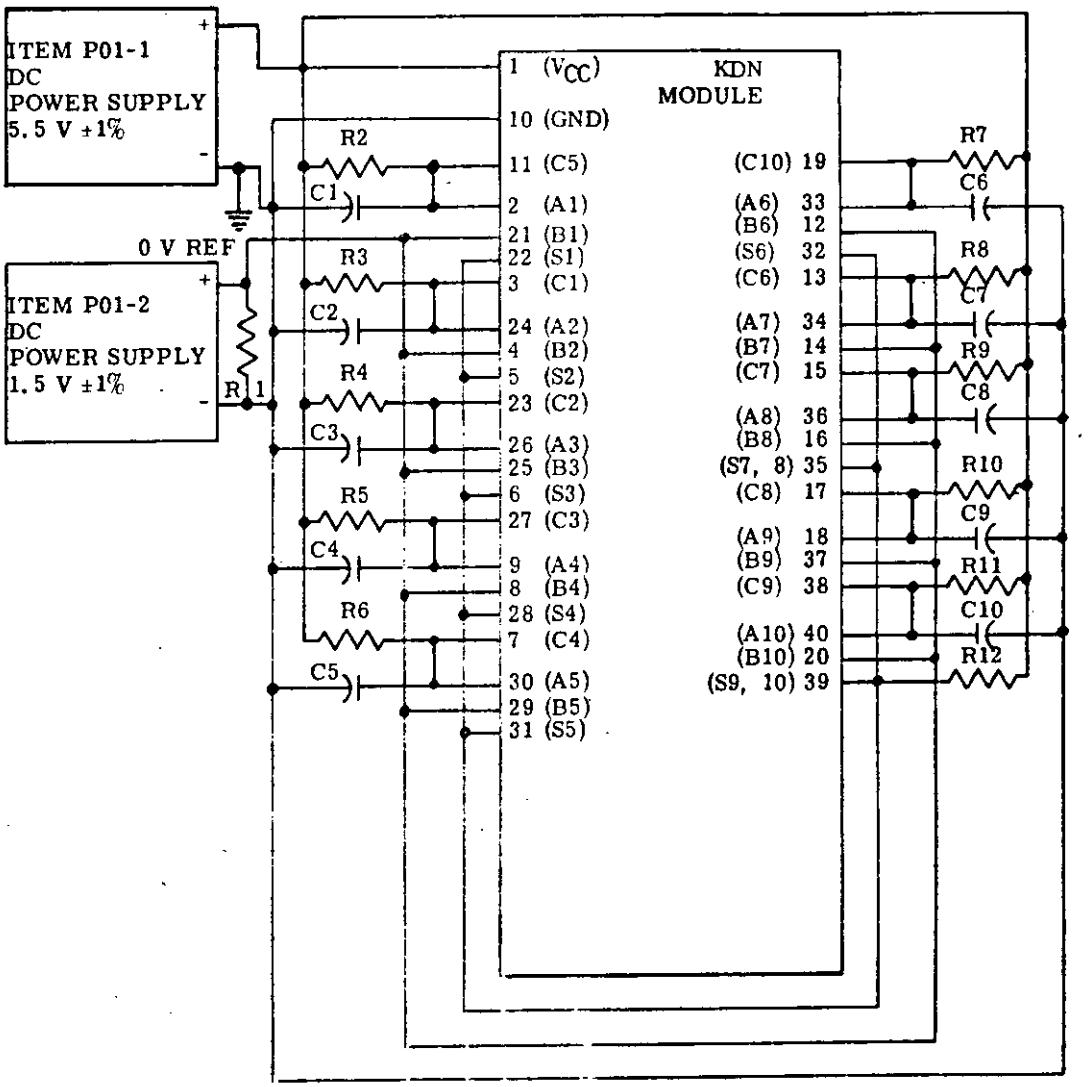
6.6 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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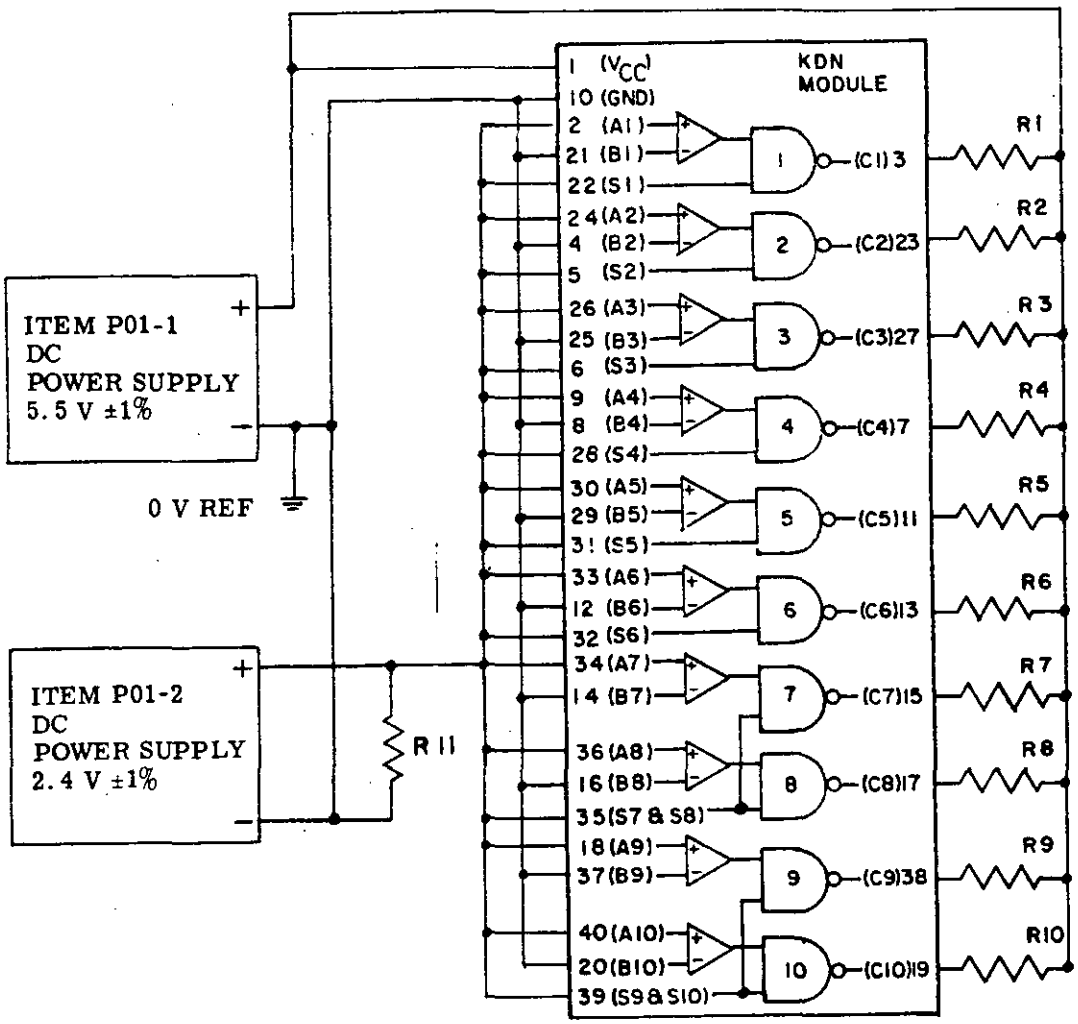
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NOTES:
1. R1 = 27 Ω ±5%, 1/4 W.
2. R2-R11 = 3.8 Ω ±1%, 1/4 W.
3. R12 = 1 kΩ ±5%, 1/4 W.
4. C1-C10 = 100 pF ±20%.

FIGURE 3 Operating temperature and life test circuit.

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NOTES:
1. R1-R10 = 348Ω ±1%, 1/4 W.
2. R11 = 51Ω ±5%, 1 W.

FIGURE 4. Transient thermal test circuit.

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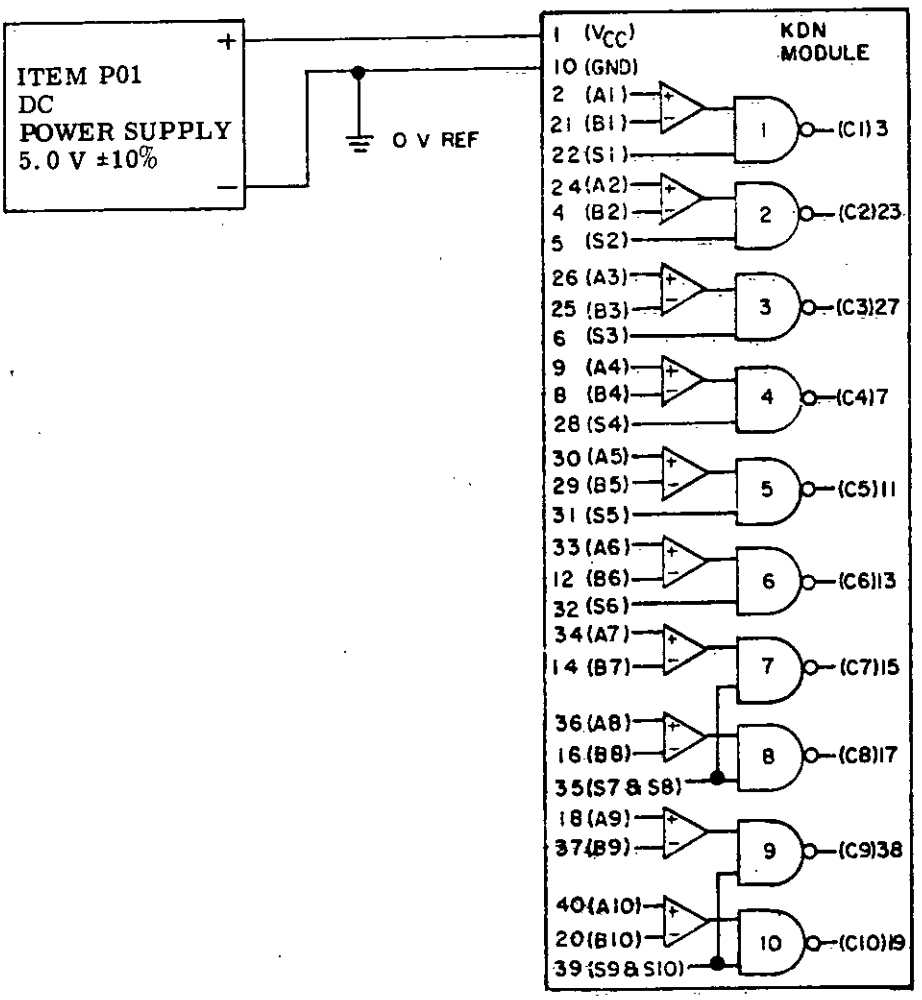


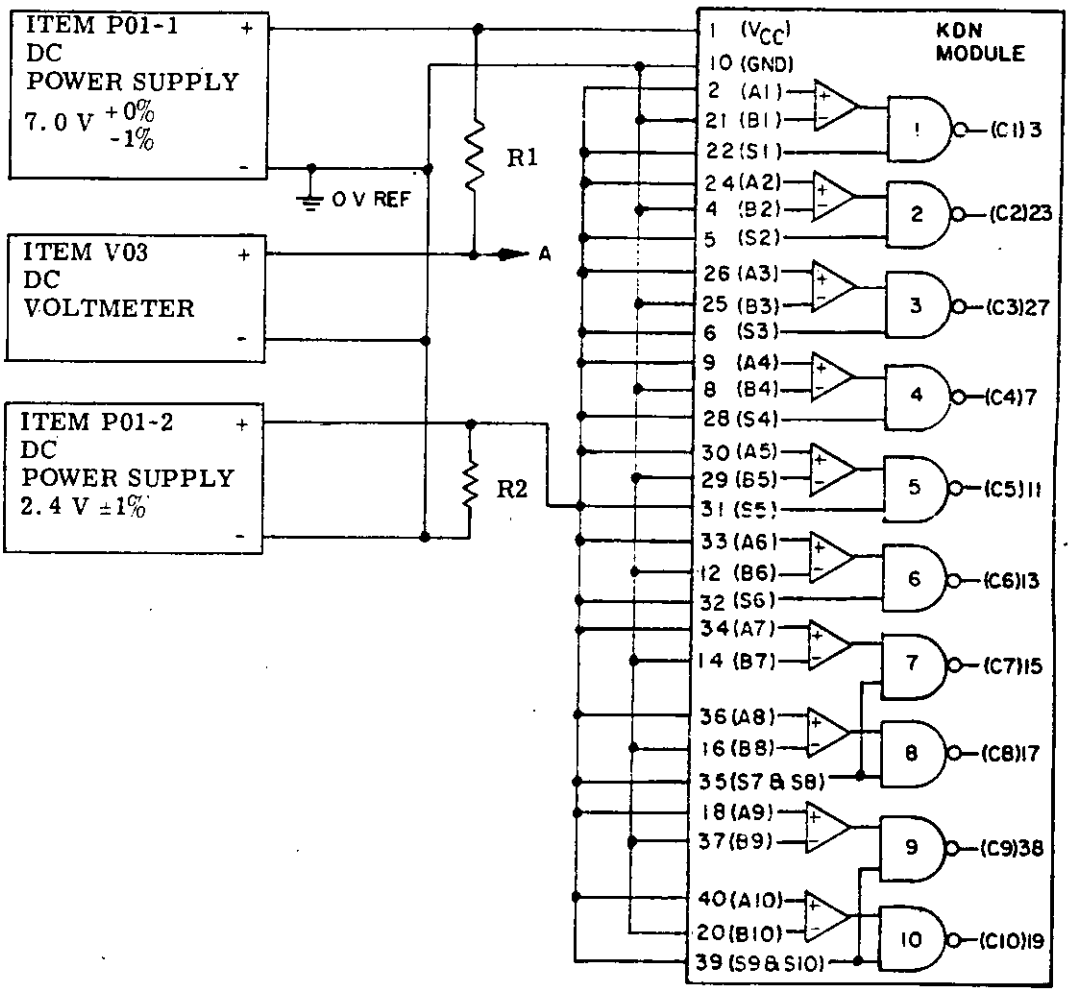
FIGURE 5. Vibration test circuit.

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TABLE III. V_{OUT} (low-loaded) test.

Test condition	Parameter	Connect	Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	Min	Max		
A1	1C V_{OUT} (low-loaded)	3	0.01	0.8	V	6
A2	2C	23				
A3	3C	27				
A4	4C	7				
A5	5C	11				
A6	6C	13				
A7	7C	15				
A8	8C	17				
A9	9C	38				
A10	10C V_{OUT} (low-loaded)	19	0.01	0.8	V	6

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NOTES:
1. R1 = 348Ω ±1%, 1/4 W.
2. R2 = 51Ω ±5%, 1 W.

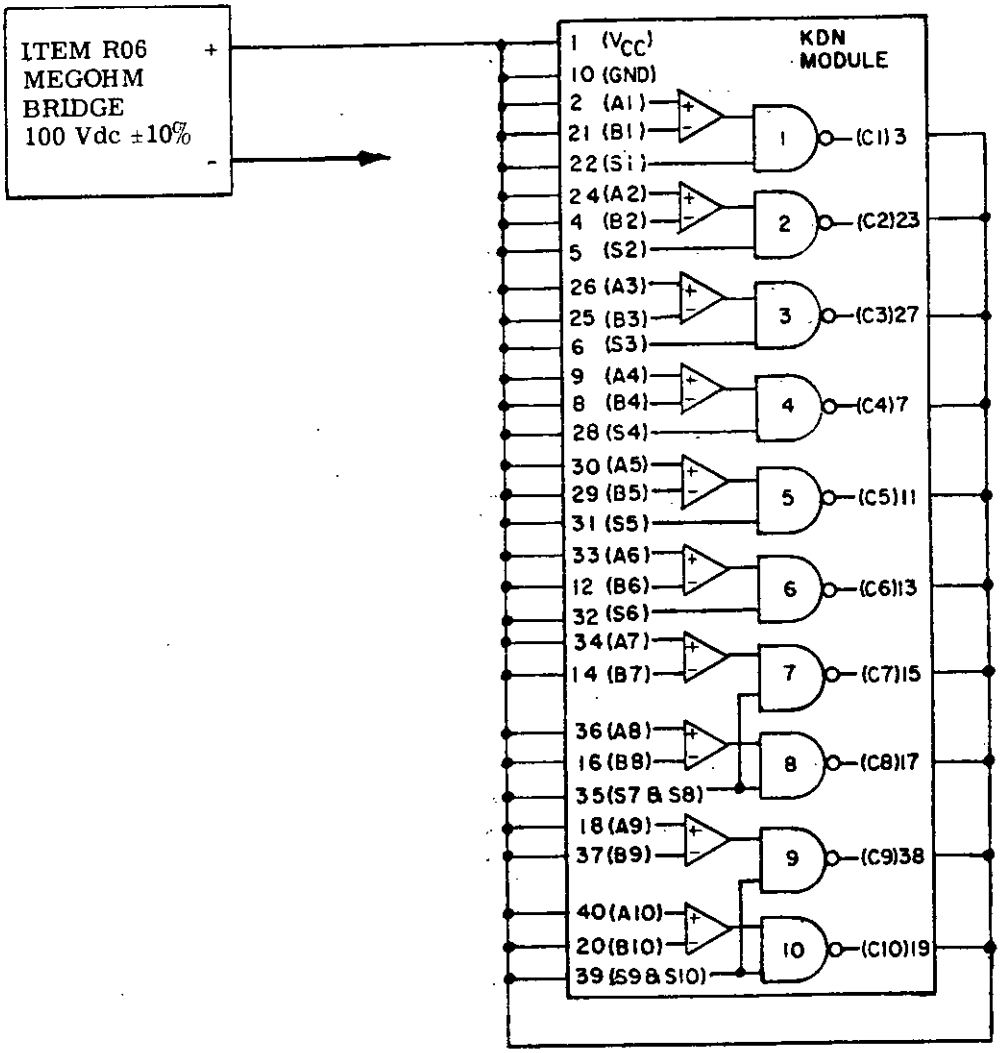
FIGURE 6. V_{OUT} (low-loaded) test circuit for table III.

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TABLE IV. Isolation test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		Positive to pin	Negative to pin	Min	Max		
B	Isolation	All pins	Module frame	10	∞	MΩ	7

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WARNING

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

NOTE: Penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

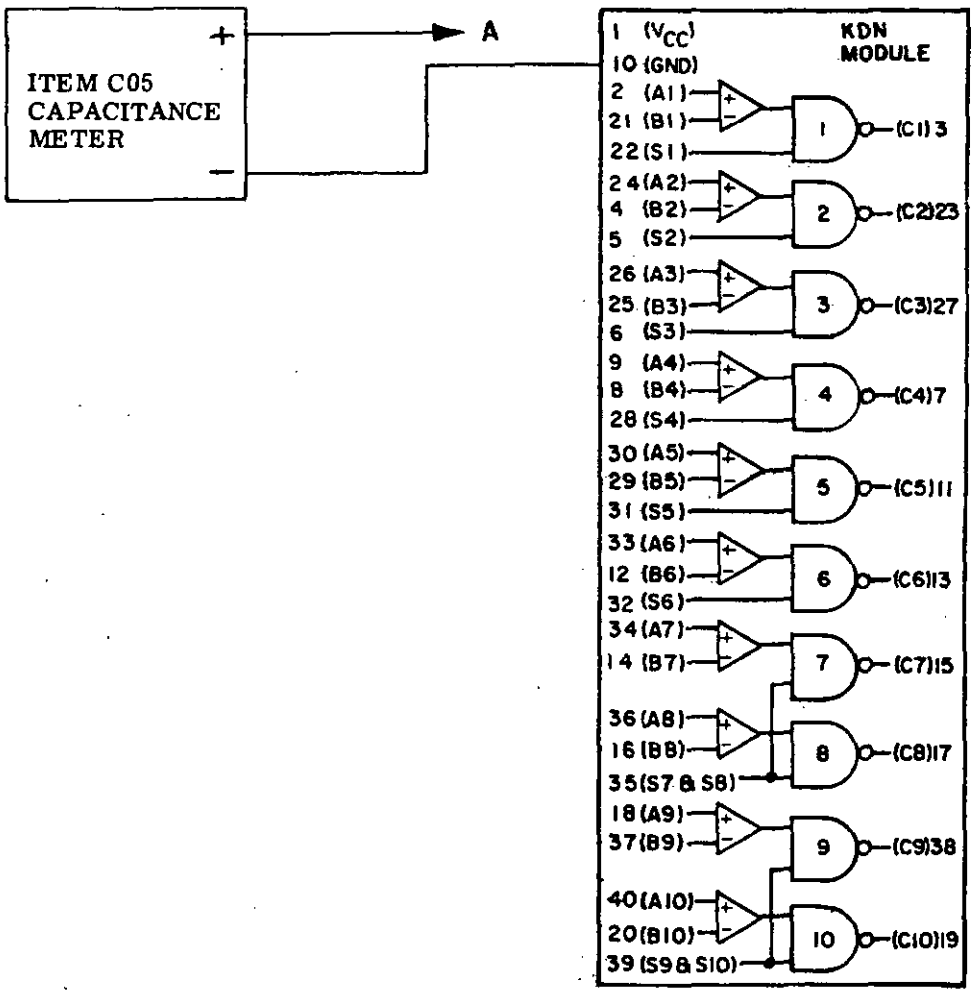
FIGURE 7. Isolation test circuit for table IV.

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TABLE V. Input capacitance test.

Test condition	Parameter		Connect	Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	Min	Max		
C1	1A	Input capacitance	2	2	20	pF	8
C2	1B		21				
C3	1S		22				
C4	2A		24				
C5	2B		4				
C6	2S		5				
C7	3A		26				
C8	3B		25				
C9	3S		6				
C10	4A		9				
C11	4B		8				
C12	4S		28				
C13	5A		30				
C14	5B		29				
C15	5S		31				
C16	6A		33				
C17	6B		12				
C18	6S		32				
C19	7A		34				
C20	7B		14				
C21	8A		36				
C22	8B		16				
C23	7S, 8S		35				
C24	9A		18				
C25	9B		37				
C26	10A		40				
C27	10B		20				
C28	9S, 10S	Input capacitance	39	2	20	pF	8

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NOTES:
1. Null out the effect of test leads prior to connecting to module.
2. Do not attach test connector or wires to any of the module pins other than pin 10 and the pin under test. Stray capacitance added to other pins may affect the readings.

FIGURE 8. Input capacitance test circuit for table V.

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TABLE VI. Static output voltage (low-level) test.

Test condition	Parameter	V_{in}	Connect								Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	G to pin	H to pin	Min	Max		
D1	1C Static output voltage, low level	2.0	F	C, G	B, G	Open	Open	A	B, C	3	0.0	0.4	V	9
D2	2C									23				
D3	3C									27				
D4	4C									7				
D5	5C									11				
D6	6C									13				
D7	7C									15				
D8	8C									17				
D9	9C									38				
D10	10C			C, G	B, G	Open	Open		B, C	19				
D11	1C			D, G	E	B, G	C		B, D	3				
D12	2C									23				
D13	3C									27				
D14	4C									7				
D15	5C									11				
D16	6C									13				
D17	7C									15				
D18	8C									17				
D19	9C									38				
D20	10C		F	D, G	E	B, G	C	A	B, D	19				
D21	1C		E, F	G	D	C	A, F	A, E	B	3				
D22	2C									23				
D23	3C									27				
D24	4C									7				
D25	5C									11				
D26	6C									13				
D27	7C									15				
D28	8C									17				
D29	9C									38				
D30	10C Static output voltage, low level	2.0	E, F	G	D	C	A, F	A, E	B	19	0.0	0.4	V	9

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TABLE VI. Static output voltage (low-level) test - Continued.

Test condition	Parameter	V_{in}	Connect								Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	G to pin	H to pin	Min	Max		
D31	1C Static output voltage, low level	2.0	F	D,G	E	B,G	C	A	B,D	3	0.0	0.4	V	9
D32	2C									23				
D33	3C									27				
D34	4C									7				
D35	5C									11				
D36	6C									13				
D37	7C									15				
D38	8C									17				
D39	9C									38				
D40	10C		F	D,G	E	B,G	C	A	B,D	19				
D41	1C		E,F	G	D	C	A,F	A,E	B	3				
D42	2C									23				
D43	3C									27				
D44	4C									7				
D45	5C									11				
D46	6C									13				
D47	7C									15				
D48	8C									17				
D49	9C									38				
D50	10C Static output voltage, low level	2.0	E,F	G	D	C	A,F	A,E	B	19	0.0	0.4	V	9

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TABLE VII. Static output voltage (high-level) test.

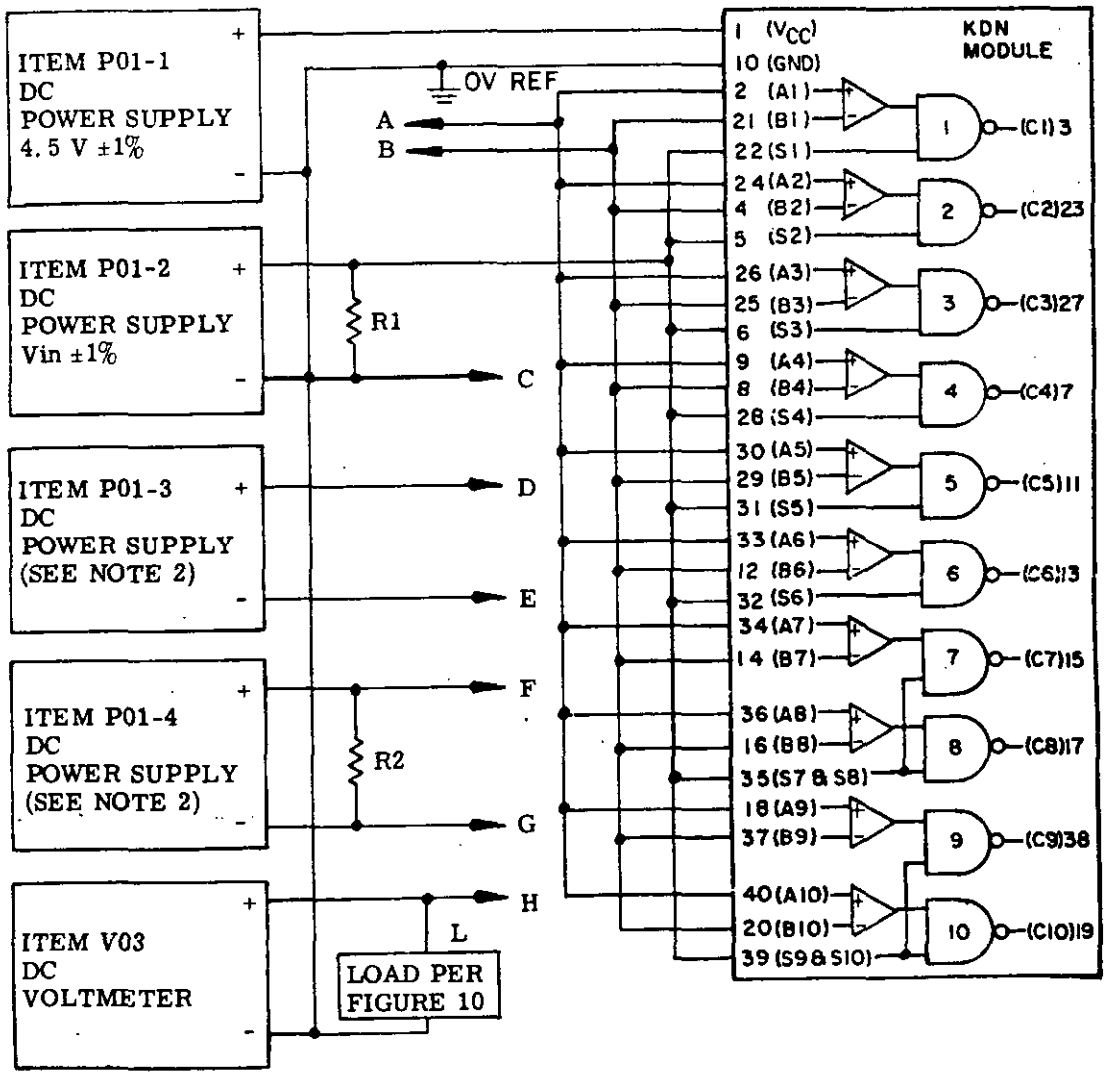
Test condition	Parameter	V_{in}	Connect								Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	G to pin	H to pin	Min	Max		
E1	1C Static output voltage, high level	0.8	F	C, G	B, G	Open	Open	A	B, C	3	2.4	4.5	V	9
E2	2C									23				
E3	3C									27				
E4	4C									7				
E5	5C									11				
E6	6C									13				
E7	7C									15				
E8	8C									17				
E9	9C									38				
E10	10C	0.8	F	C, G	B, G			A	B, C	19				
E11	1C	2.0	C, G	F	A, G			B	A, C	3				
E12	2C									23				
E13	3C									27				
E14	4C									7				
E15	5C									11				
E16	6C									13				
E17	7C									15				
E18	8C									17				
E19	9C									38				
E20	10C		C, G		A, G	Open	Open		A, C	19				
E21	1C		D, G		E	A, G	C		A, D	3				
E22	2C									23				
E23	3C									27				
E24	4C									7				
E25	5C									11				
E26	6C									13				
E27	7C									15				
E28	8C									17				
E29	9C									38				
E30	10C		D, G	F	E	A, G	C	B	A, D	19				
E31	1C		G	E, F	D	C	B, F	B, E	A	3				
E32	2C									23				
E33	3C									27				
E34	4C									7				
E35	5C Static output voltage, high level	2.0	G	E, F	D	C	B, F	B, E	A	11	2.4	4.5	V	9

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TABLE VII. Static output voltage (high-level) test - Continued.

Test condition	Parameter	V_{in}	Connect								Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	G to pin	H to pin	Min	Max		
E36	6C Static output voltage, high level	2.0	G	E,F	D	C	B,F	B,E	A	13	2.4	4.5	V	9
E37	7C									15				
E38	8C									17				
E39	9C									38				
E40	10C		G	E,F	D	C	B,F	B,E	A	19				
E41	1C		D,G	F	E	A,G	C	B	A,D	3				
E42	2C									23				
E43	3C									27				
E44	4C									7				
E45	5C									11				
E46	6C									13				
E47	7C									15				
E48	8C									17				
E49	9C									38				
E50	10C		D,G	F	E	A,G	C	B	A,D	19				
E51	1C		G	E,F	D	C	B,F	B,E	A	3				
E52	2C									23				
E53	3C									27				
E54	4C									7				
E55	5C									11				
E56	6C									13				
E5	7C									15				
E58	8C									17				
E59	9C									38				
E60	10C Static output voltage, high level	2.0	G	E,F	D	C	B,F	B,E	A	19	2.4	4.5	V	9

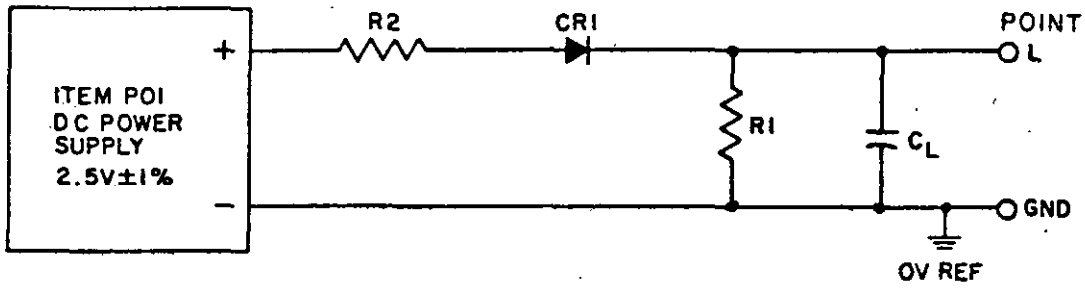
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NOTES:
1. R1 = R2 = 51Ω ±5%, 1 W.
2. For test conditions D1-D10 and E1-E20:
   Item P01-4 = 0.75 V ±1%.
   For test conditions D11-D30 and E41-E60:
   Item P01-3 = 7.0 V ±1%
   Item P01-4 = 0.75 V ±1%.
   For test conditions D31-D50 and E21-E40:
   Item P01-3 = 13.9 V ±1%
   Item P01-4 = 1.0 V ±1%.

FIGURE 9. Static output voltage test circuit for table VI (low-level) and table VII (high-level).

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NOTES:
1. C_L = 100 ±5 pF (including probe and parasitic capacitance) for test conditions N, P, R, S, T, and U.
2. CR1 = 1N4150 or equivalent.
3. R1 = 487Ω ±1%, for test conditions D and E.
4. R1 = 2.67 kΩ ±1%, for test conditions N, P, R, S, T, U, V, W, Z, and AA.
5. R2 = Select value for a current flow of 16.2 mA ±1% from point L to ground.
6. Capacitor C_L is not required for dc electrical test conditions D and E.
7. C_L = 50 ±5 pF (including probe and parasitic capacitance) for test conditions V, W, Z, and AA.
8. The maximum lead length from point L to the appropriate module pin shall be 3 inches (76.2 mm).
9. During output switching, the 2.5 V power supply shall have a maximum overshoot of 500 mV.

FIGURE 10. Logic load circuit.

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TABLE VIII. Input current, +15 V (differential inputs), test.

Test condition	Parameter	V_{CC}	V_{in}	Connect							Initial limits 25° ±5°C Ambient		Unit	Test figure
				A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	G to pin	Min	Max		
F1	1A Input current, +15 V (differential inputs)	5.5	15.0	Open	2	E	21, G	C	Open	21, D	2.0	6.0	mA	11
F2	1B				21		2, G			2, D				
F3	2A				24		4, G			4, D				
F4	2B				4		24, G			24, D				
F5	3A				26		25, G			25, D				
F6	3B				25		26, G			26, D				
F7	4A				9		8, G			8, D				
F8	4B				8		9, G			9, D				
F9	5A				30		29, G			29, D				
F10	5B				29		30, G			30, D				
F11	6A				33		12, G			12, D				
F12	6B				12		33, G			33, D				
F13	7A				34		14, G			14, D				
F14	7B				14		34, G			34, D				
F15	8A				36		16, G			16, D				
F16	8B				16		36, G			36, D				
F17	9A				18		37, G			37, D				
F18	9B				37		18, G			18, D				
F19	10A				40		20, G			20, D				
F20	10B Input current, +15 V (differential inputs)	5.5	15.0	Open	20	E	40, G	C	Open	40, D	2.0	6.0	mA	11

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TABLE IX. Input current, -15 V (differential inputs), test.

Test condition	Parameter	V_{CC}	V_{in}	Connect							Initial limits 25° ±5°C Ambient		Unit	Test figure
				A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	G to pin	Min	Max		
G1	1A Input current, -15 V (differential inputs)	5.5	15.0	Open	D	2	B	21, G	Open	21, E	2.6	7.8	mA	11
G2	1B					21		2, G		2, E				
G3	2A					24		4, G		4, E				
G4	2B					4		24, G		24, E				
G5	3A					26		25, G		25, E				
G6	3B					25		26, G		26, E				
G7	4A					9		8, G		8, E				
G8	4B					8		9, G		9, E				
G9	5A					30		29, G		29, E				
G10	5B					29		30, G		30, E				
G11	6A					33		12, G		12, E				
G12	6B					12		33, G		33, E				
G13	7A					34		14, G		14, E				
G14	7B					14		34, G		34, E				
G15	8A					36		16, G		16, E				
G16	8B					16		36, G		36, E				
G17	9A					18		37, G		37, E				
G18	9B					37		18, G		18, E				
G19	10A					40		20, G		20, E				
G20	10B Input current, -15 V (differential inputs)	5.5	15.0	Open	D	20	B	40, G	Open	40, E	2.6	7.8	mA	11

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TABLE X. Input current, low level (differential inputs), test.

Test condition	Parameter	V_{CC}	V_{in}	Connect							Initial limits 25° ±5°C Ambient		Unit	Test figure
				A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	G to pin	Min	Max		
H1	1A Input current, low level (differential inputs)	5.5	N/A	21	Open	Open	Open	Open	2	Open	0.1	0.9	mA	11
H2	1B			2					21					
H3	2A			4					24					
H4	2B			24					4					
H5	3A			25					26					
H6	3B			26					25					
H7	4A			8					9					
H8	4B			9					8					
H9	5A			29					30					
H10	5B			30					29					
H11	6A			12					33					
H12	6B			33					12					
H13	7A			14					34					
H14	7B			34					14					
H15	8A			16					36					
H16	8B			36					16					
H17	9A			37					18					
H18	9B			18					37					
H19	10A			20					40					
H20	10B Input current, low level (differential inputs)	5.5	N/A	40	Open	Open	Open	Open	20	Open	0.1	0.9	mA	11

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TABLE XI. Input current, low level (strobe and common strobe), test.

Test condition	Parameter	V_{CC}	V_{in}	Connect							Initial limits 25° ±5°C Ambient		Unit	Test figure
				A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	G to pin	Min	Max		
J1	1S Input current, low level (strobe)	5.5	2.4	Open	2	7, G	Open	Open	22	21, C	0.5	2.6	mA	11
J2	2S				24	4, G			5	4, C				
J3	3S				26	25, G			6	25, C				
J4	4S				9	8, G			28	8, C				
J5	5S				30	29, G			31	29, C				
J6	6S Input current, low level (strobe)				33	12, G			32	12, C	0.5	2.6		
J7	7S, 8S Input current, low level (common strobe)				34, 36	14, 16, G			35	14, 16, C	1.0	5.2		
J8	9S, 10S Input current, low level (common strobe)	5.5	2.4	Open	18, 40	37, 20, G	Open	Open	39	37, 20, C	1.0	5.2	mA	11

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TABLE XII. Input current, high level (strobe and common strobe), test.

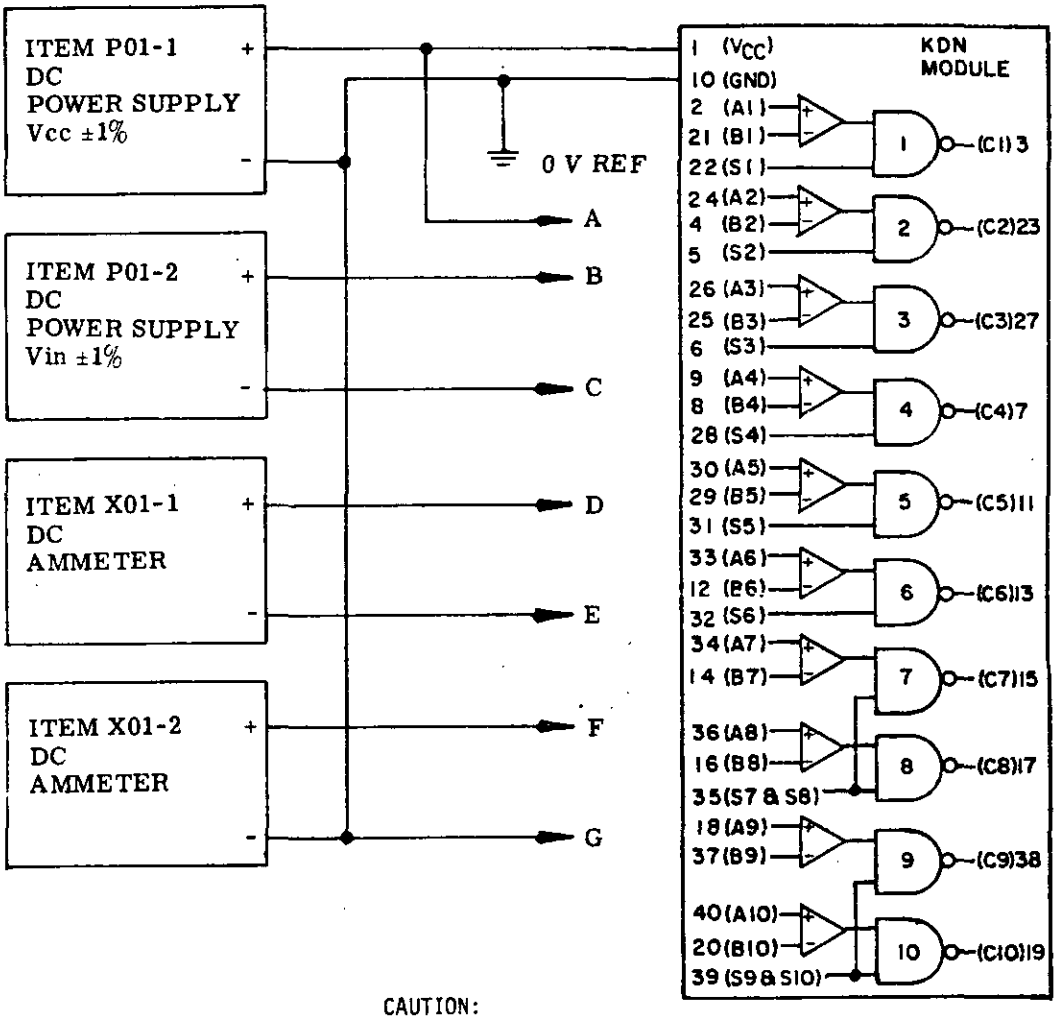
Test condition	Parameter	V_{CC}	V_{in}	Connect							Initial limits 25° ±5°C Ambient		Unit	Test figure
				A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	G to pin	Min	Max		
K1	1S Input current, high level (strobe)	4.5	2.4	D	21	2, G	A	22	Open	2, C	0.0	2.0	µA	11
K2	2S				4	24, G		5		24, C				
K3	3S				25	26, G		6		26, C				
K4	4S				8	9, G		28		9, C				
K5	5S				29	30, G		31		30, C				
K6	6S Input current, high level (strobe)				12	33, G		32		33, C		2.0		
K7	7S, 8S Input current, high level (common strobe)				14, 16	34, 36, G		35		34, 36, C		4.0		
K8	9S, 10S Input current, high level (common strobe)	4.5	2.4	D	37, 20	18, 40, G	A	39	Open	18, 40, C	0.0	4.0	µA	11

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TABLE XIII. Output short circuit current test.

Test condition	Parameter	V_{CC}	V_{in}	Connect							Initial limits 25° ±5°C Ambient		Unit	Test figure
				A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	G to pin	Min	Max		
1.1	1C Output short circuit current	5.5	2.4	Open	21	[illegible], G	Open	Open	3	2, C	15	80	mA	11
1.2	2C				4	24, G			23	24, C				
1.3	3C				25	26, G			27	26, C				
1.4	4C				8	9, G			7	9, C				
1.5	5C				29	30, G			11	30, C				
1.6	6C				12	33, G			13	33, C				
1.7	7C				14	34, G			15	34, C				
1.8	8C				16	36, G			17	36, C				
1.9	9C				37	18, G			38	18, C				
1.10	10C Output short circuit current	5.5	2.4	Open	20	40, G	Open	Open	19	40, C	15	80	mA	11

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CAUTION:

Do not isolate the chassis of ammeter XO1 from earth ground.

NOTES:
1. Since the ammeters are not used simultaneously, only one dc ammeter (XO1) is required. Use ammeter XO1-2 for tables IX, X, and XII; use ammeter XO1-1 for tables VII, VIII, and XI.
2. With point "A" disconnected, ammeter XO1-1 should indicate zero current on the 1 µA range.
3. When performing output short circuit tests, only one output shall be shorted at a time and the measurement taken within 5 seconds.

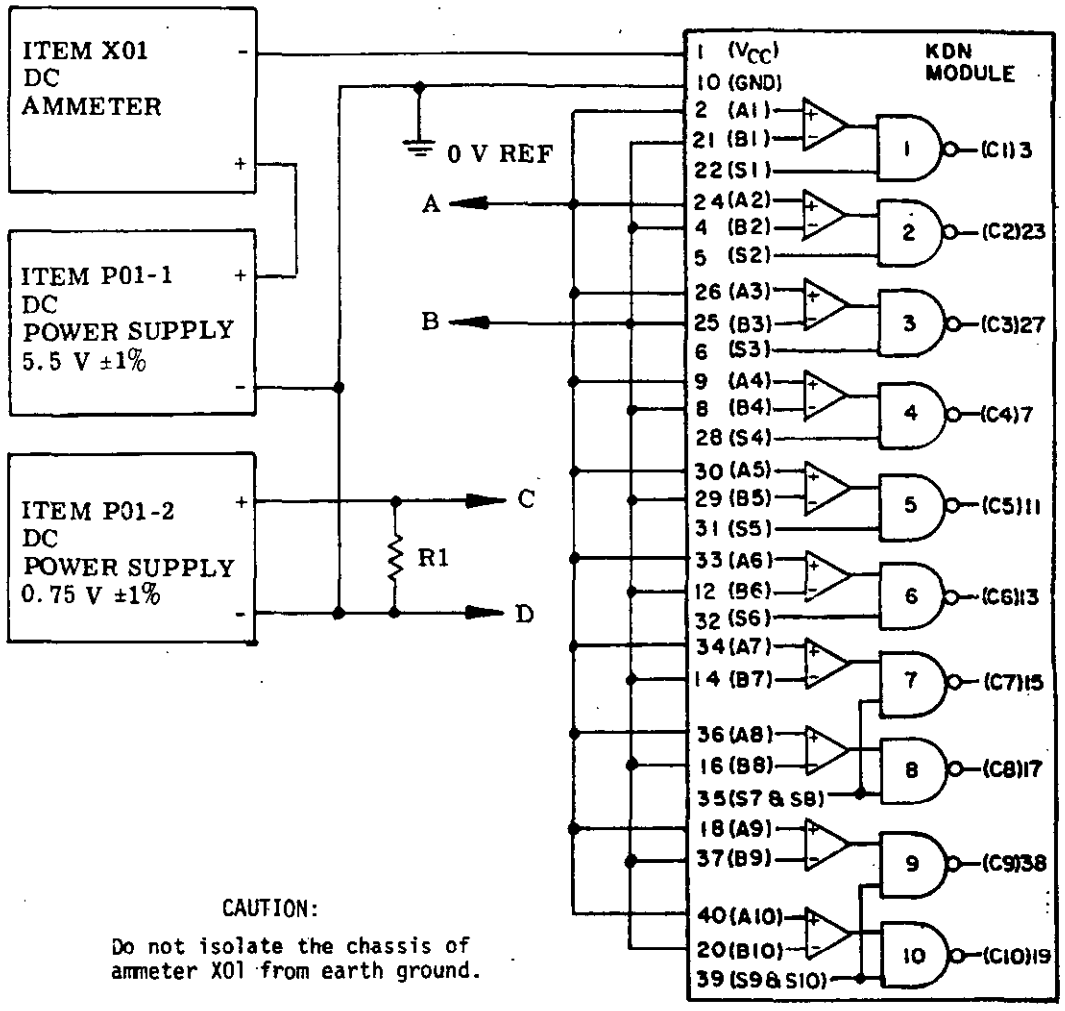
FIGURE 11. Input current and output short circuit current test circuit for tables VIII, IX, X, XI, XII, and XIII.

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TABLE XIV. Power supply current (low-level and high-level) test.

Test condition	Parameter	Connect				Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
M1	Power supply current, low level	C	D	A	B	1	250	mA	12
M2	Power supply current, high level	D	C	B	A	1	250	mA	12

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NOTE: R1 = 51Ω ±5%, 1 W.

FIGURE 12. Power supply current test circuit for table XIV.

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TABLE XV. Transition time (t_{TLH}, t_{THL}) and propagation delay time (t_{PHL}, t_{PLH}) test.

Test condition	Parameter	Connect					Initial limits 25° ±5°C Ambient				Initial limits 25° ±5°C Ambient				Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	t_{TLH}		t_{THL}		t_{PHL}		t_{PLH}			
							Min	Max	Min	Max	Min	Max	Min	Max		
N1, P1, R1, T1	1C "A" input	2	21	22	Open	3	4	35	4	15	10	53	10	57	ns	13,14
N2, P2, R2, T2	1C "B" input	21	2	22		3										
N3, P3, R3, T3	2C "A" input	24	4	5		23										
N4, P4, R4, T4	2C "B" input	4	24	5		23										
N5, P5, R5, T5	3C "A" input	26	25	6		27										
N6, P6, R6, T6	3C "B" input	25	26	6		27										
N7, P7, R7, T7	4C "A" input	9	8	28		7										
N8, P8, R8, T8	4C "B" input	8	9	28		7										
N9, P9, R9, T9	5C "A" input	30	29	31		11										
N10, P10, R10, T10	5C "B" input	29	30	31		11										
N11, P11, R11, T11	6C "A" input	33	12	32		13										
N12, P12, R12, T12	6C "B" input	12	33	32		13										
N13, P13, R13, T13	7C "A" input	34	14	35		15										
N14, P14, R14, T14	7C "B" input	14	34	35		15										
N15, P15, R15, T15	8C "A" input	36	16	35		17										
N16, P16, R16, T16	8C "B" input	16	36	35		17										
N17, P17, R17, T17	9C "A" input	18	37	39		38										
N18, P18, R18, T18	9C "B" input	37	18	39	Open	38	4	35	4	15	10	53	10	57	ns	13,14

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TABLE XV. Transition time (t_{TLH}, t_{THL}) and propagation delay time (t_{PHL}, t_{PLH}) test - Continued.

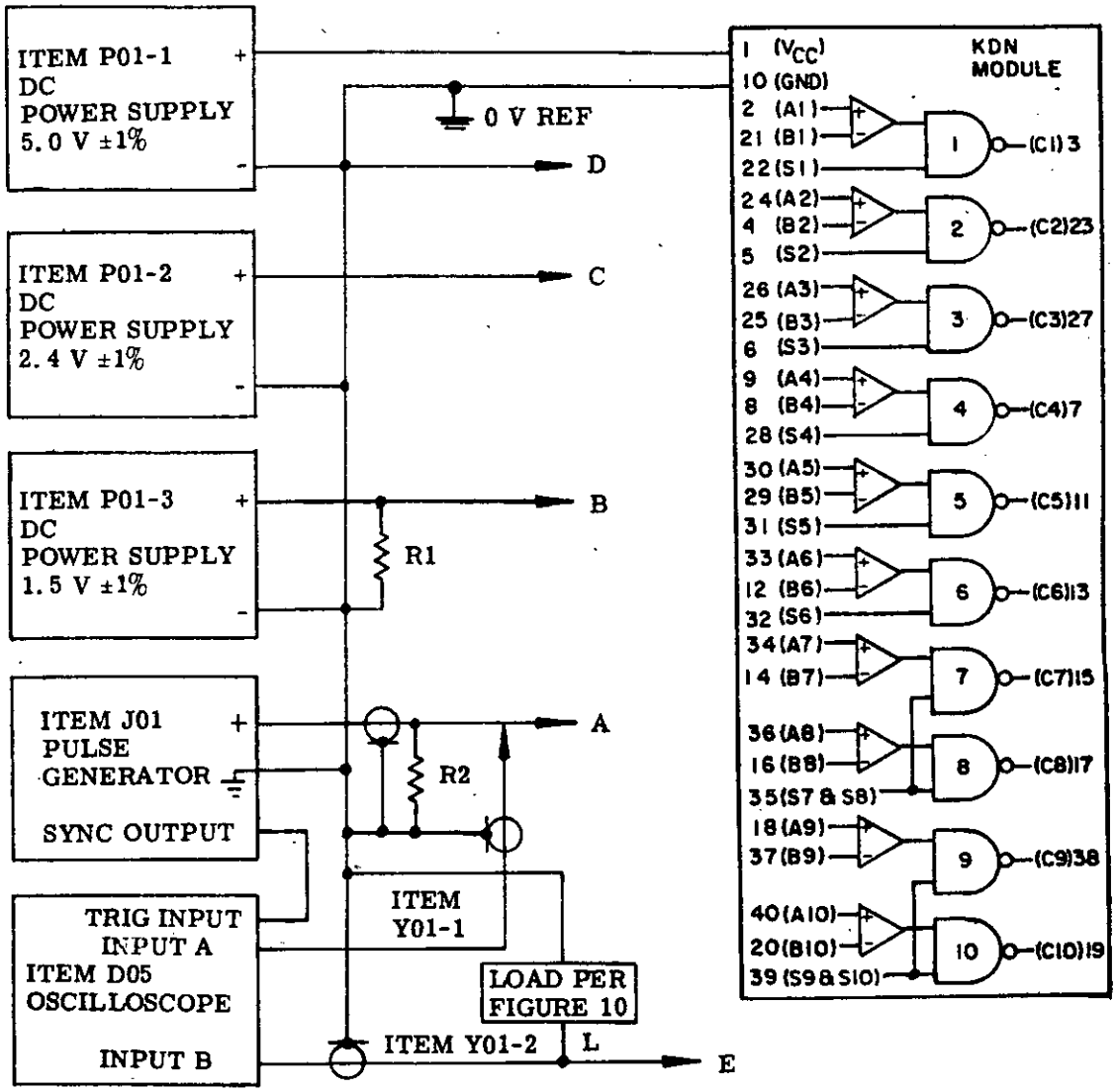
Test condition	Parameter	Connect					Initial limits 25° ±5°C Ambient				Initial limits 25° ±5°C Ambient				Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	t_{TLH}		t_{THL}		t_{PHL}		t_{PLH}			
							Min	Max	Min	Max	Min	Max	Min	Max		
N19, P19, R19, T19 N20, P20, R20, T20	10C "A" input	40	20	39	Open	19	4	35	4	15	10	53	10	57	ns	13,14
	10C "B" input	20	40	39	Open	19	4	35	4	15	10	53	10	57	ns	13,14

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TABLE XVI. Propagation delay time (t_{PHL}, t_{PLH}) test.

	Test condition	Parameter	Connect					Initial limits 25° ±5°C Ambient				Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	t_{PHL}		t_{PLH}			
								Min	Max	Min	Max		
*	S1 & U1	1C "S" input	22	Open	2	21	3	5	29	5	26	ns	13,15
	S2 & U2	2C	5		24	4	23						
	S3 & U3	3C	6		26	25	27						
	S4 & U4	4C	28		9	8	7						
	S5 & U5	5C	31		30	29	11						
	S6 & U6	6C	32		33	12	13						
	S7 & U7	7C	35		34	14	15						
	S8 & U8	8C	35		36	16	17						
	S9 & U9	9C	39		18	37	38						
*	S10 & U10	10C "S" input	39	Open	40	20	19	5	29	5	26	ns	13,15

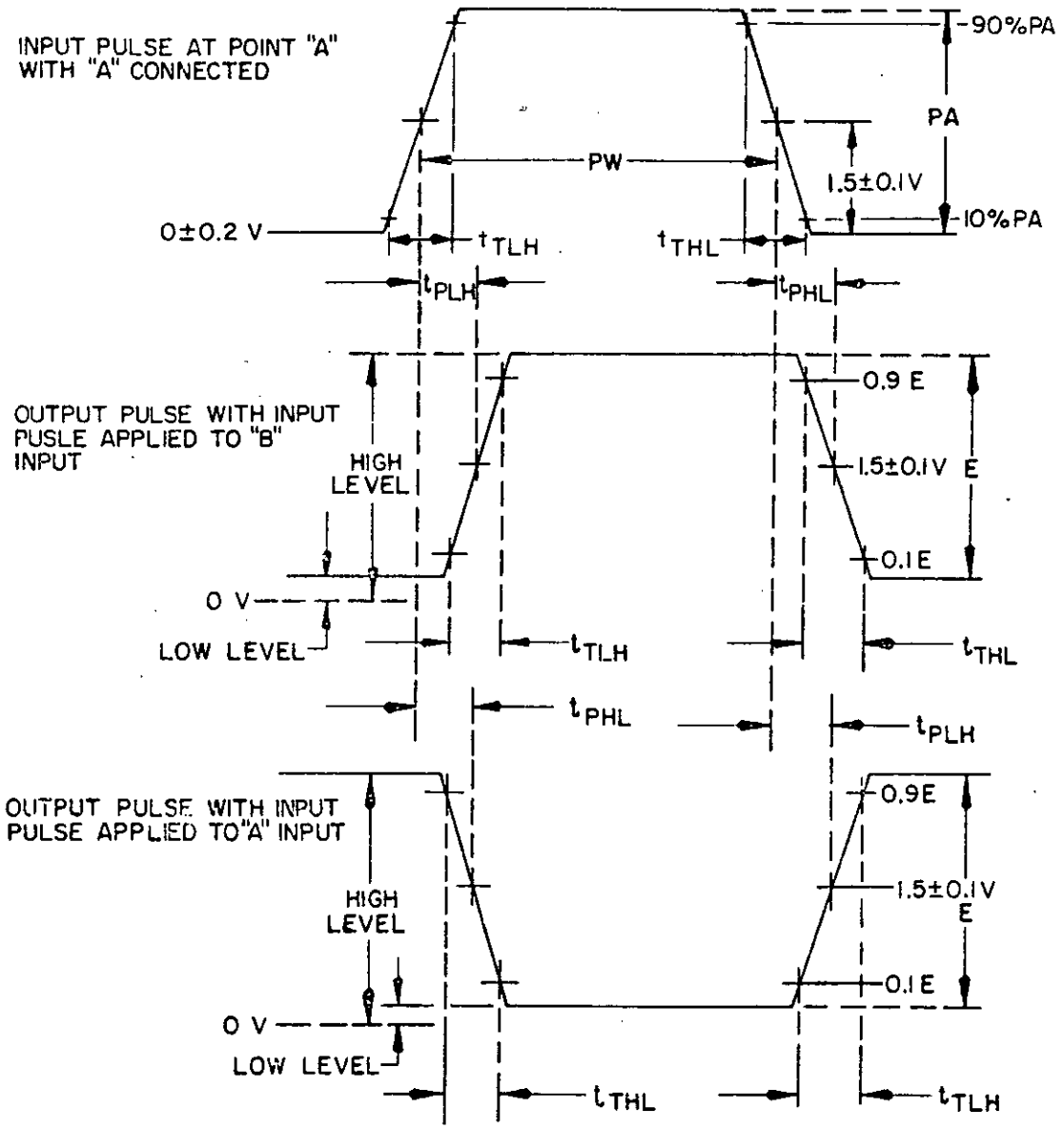
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NOTES:
1. R1 = R2 = 51Ω ±5%, 1 W.
2. Before making these measurements, the output levels shall be verified to be 0.4 V maximum for the low level and 2.4 V minimum for the high level.

FIGURE 13. Transition time and propagation delay time test circuit for tables XV and XVI.

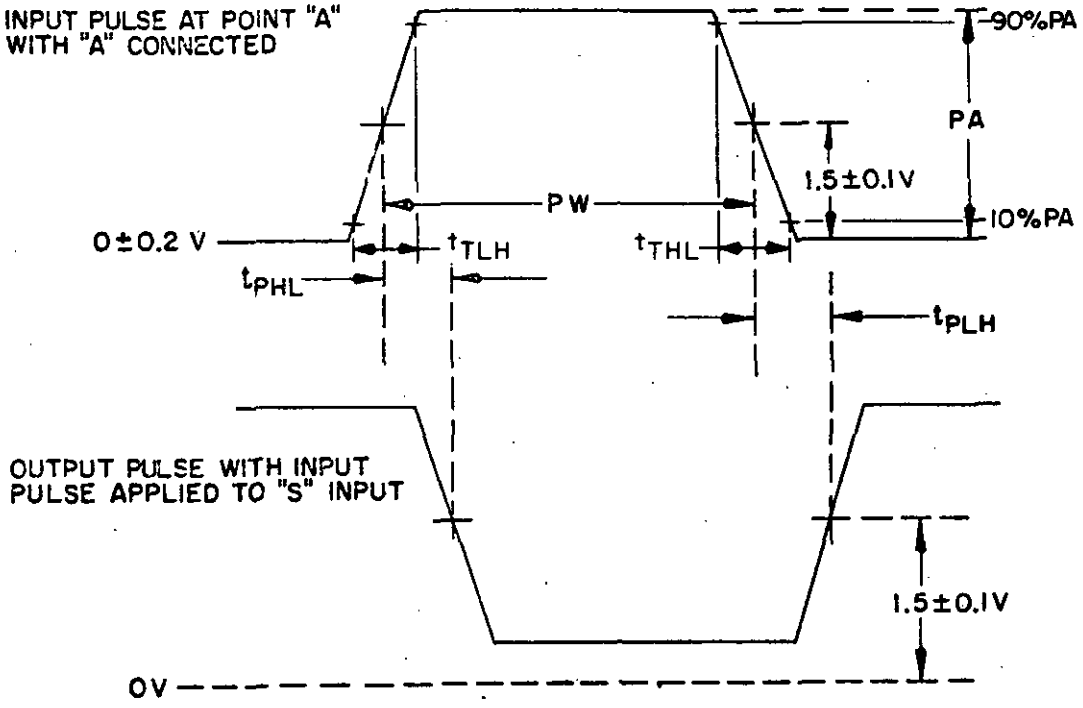
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NOTES: Input pulse parameters:
1. PA = 3.0 ±0.1 V.
2. PW = 110 ±10 ns.
3. $t_{TLH} = t_{THL} = 10 \pm 2$ ns.
4. PRR = 500 kHz ±10%.

FIGURE 14. Transition time and propagation delay time test circuit waveforms for table XV.

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NOTES: Input pulse parameters:
1. PA = 3.5 ±0.1 V.
2. PW = 110 ±10 ns.
3. $t_{TLH} = t_{THL}$ = 10 ±2 ns.
4. PRR = 500 kHz ±10%.

FIGURE 15. Propagation delay time test circuit waveforms for table XVI.

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TABLE XVII. Output crosstalk (low-level) test.

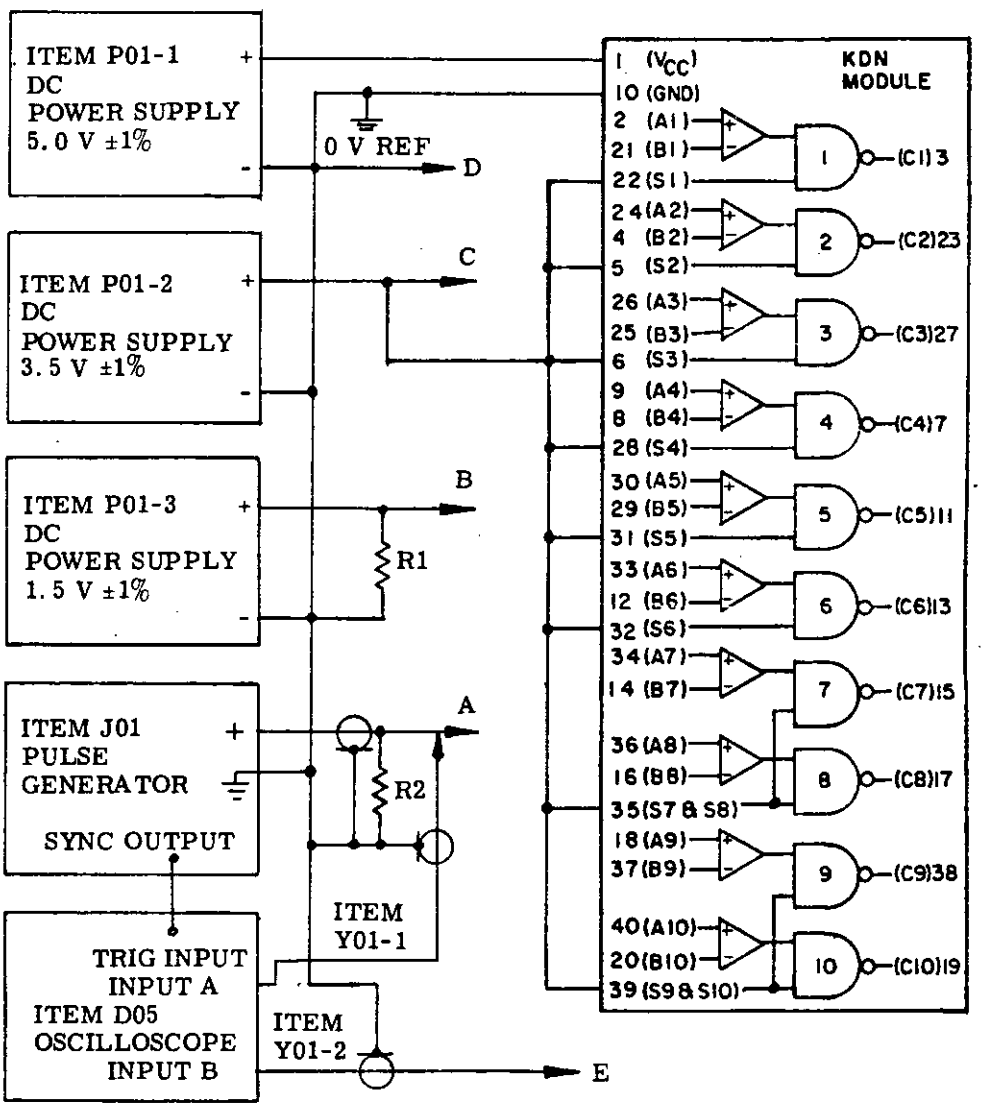
Test condition	Parameter	Connect					Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
V1	1C Output crosstalk, low level	24, 26, 9, 30, 33, 34, 36, 18, 40	4, 25, 8 29, 12, 14, 16, 37, 20	2	21	3	0	500	mV	16,17
V2	2C	2, 26, 9, 30, 33, 34, 36, 18, 40	21, 25, 8, 29, 12, 14, 16, 37, 20	24	4	23				
V3	3C	2, 24, 9, 30, 33, 34, 36, 18, 40	21, 4, 8, 29, 12, 14, 16, 37, 20	26	25	27				
V4	4C	2, 24, 26, 30, 33, 34, 36, 18, 40	21, 4, 25, 29, 12, 14, 16, 37, 20	9	8	7				
V5	5C	2, 24, 26, 9, 33, 34, 36, 18, 40	21, 4, 25, 8, 12, 14, 16, 37, 20	30	29	11				
V6	6C	2, 24, 26, 9, 30, 34, 36, 18, 40	21, 4, 25, 8, 29, 14, 16, 37, 20	33	12	13				
V7	7C	2, 24, 26, 9, 30, 33, 36, 18, 40	21, 4, 25, 8, 29, 12, 16, 37, 20	34	14	15				
V8	8C	2, 24, 26, 9, 30, 33, 34, 18, 40	21, 4, 25, 8, 29, 12, 14, 37, 20	36	16	17				
V9	9C	2, 24, 26, 9, 30, 33, 34, 36, 40	21, 4, 25, 8, 29, 12, 14, 16, 20	18	37	38				
V10	10C Output crosstalk, low level	2, 24, 26, 9, 30, 33, 34, 36, 18	21, 4, 25 8, 29, 12, 14, 16, 37	40	20	19	0	500	mV	16,17

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TABLE XVIII. Output crosstalk (high-level) test.

Test condition	Parameter	Connect					Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
W1	1C Output crosstalk, high level	24, 26, 9, 30, 33, 34, 36, 18, 40	4, 25, 8, 29, 12, 14, 16, 37, 20	21	2	3	0	400	mV	16,17
W2	2C	2, 26, 9, 30, 33, 34, 36, 18, 40	21, 25, 8, 29, 12, 14, 16, 37, 20	4	24	23				
W3	3C	2, 24, 9, 30, 33, 34, 36, 18, 40	21, 4, 8, 29, 12, 14, 16, 37, 20	25	26	27				
W4	4C	2, 24, 26, 30, 33, 34, 36, 18, 40	21, 4, 25, 29, 12, 14, 16, 37, 20	8	9	7				
W5	5C	2, 24, 26, 9, 33, 34, 36, 18, 40	21, 4, 25, 8, 12, 14, 16, 37, 20	29	30	11				
W6	6C	2, 24, 26, 9, 30, 34, 36, 18, 40	21, 4, 25, 8, 29, 14, 16, 37, 20	12	33	13				
W7	7C	2, 24, 26, 9, 30, 33, 36, 18, 40	21, 4, 25, 8, 29, 12, 16, 37, 20	14	34	15				
W8	8C	2, 24, 26, 9, 30, 33, 34, 18, 40	21, 4, 25, 8, 29, 12, 14, 37, 20	16	36	17				
W9	9C	2, 24, 26, 9, 30, 33, 34, 36, 40	21, 4, 25, 8, 29, 12, 14, 16, 20	37	18	38				
W10	10C Output crosstalk, high level	2, 24, 26, 9, 30, 33, 34, 36, 18	21, 4, 25, 8, 29, 12, 14, 16, 37	20	40	19	0	400	mV	16,17

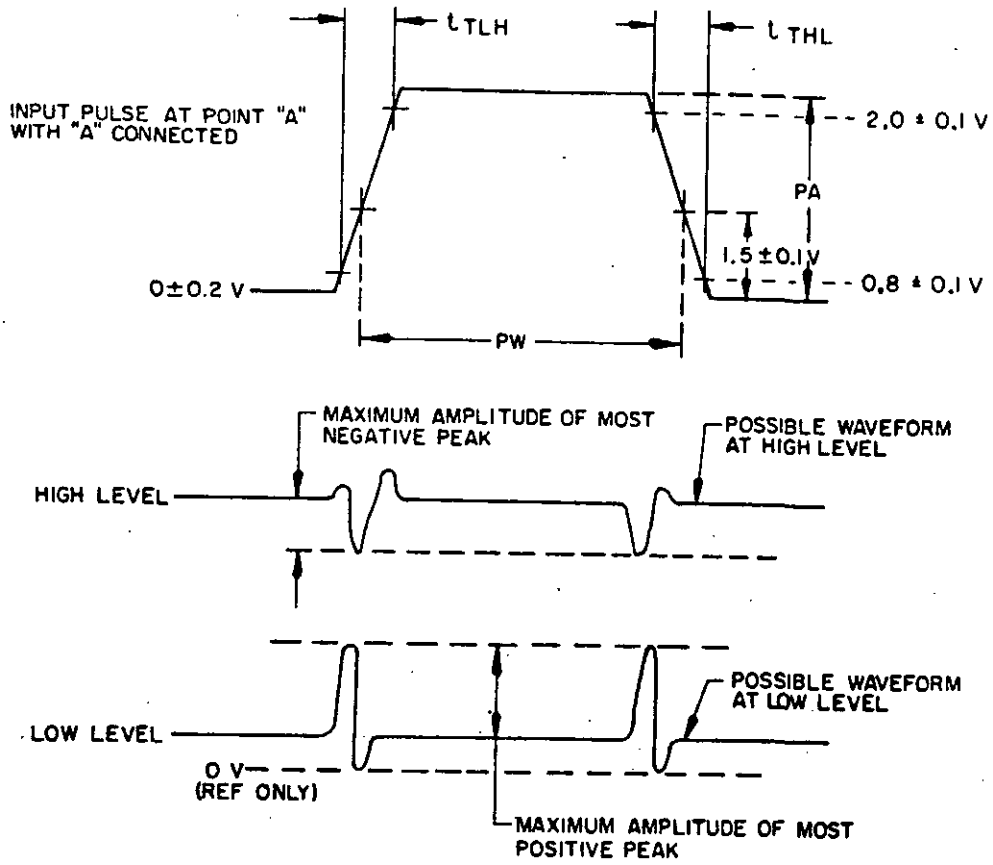
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NOTES:
1. R1 = R2 = 51Ω ±5%, 1 W.
2. Load each output with a load as specified on figure 10.

FIGURE 16. Output crosstalk test circuit for tables XVII and XVIII.

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NOTES: Input pulse parameters:
1. PW = 200 ns ±10%.
2. $t_{TLH} = t_{THL}$ = 3.5 ±1.5 ns (verified at each input pin).
3. PA = 3.5 ±0.1 V.
4. PRR = 1 MHz ±10%.

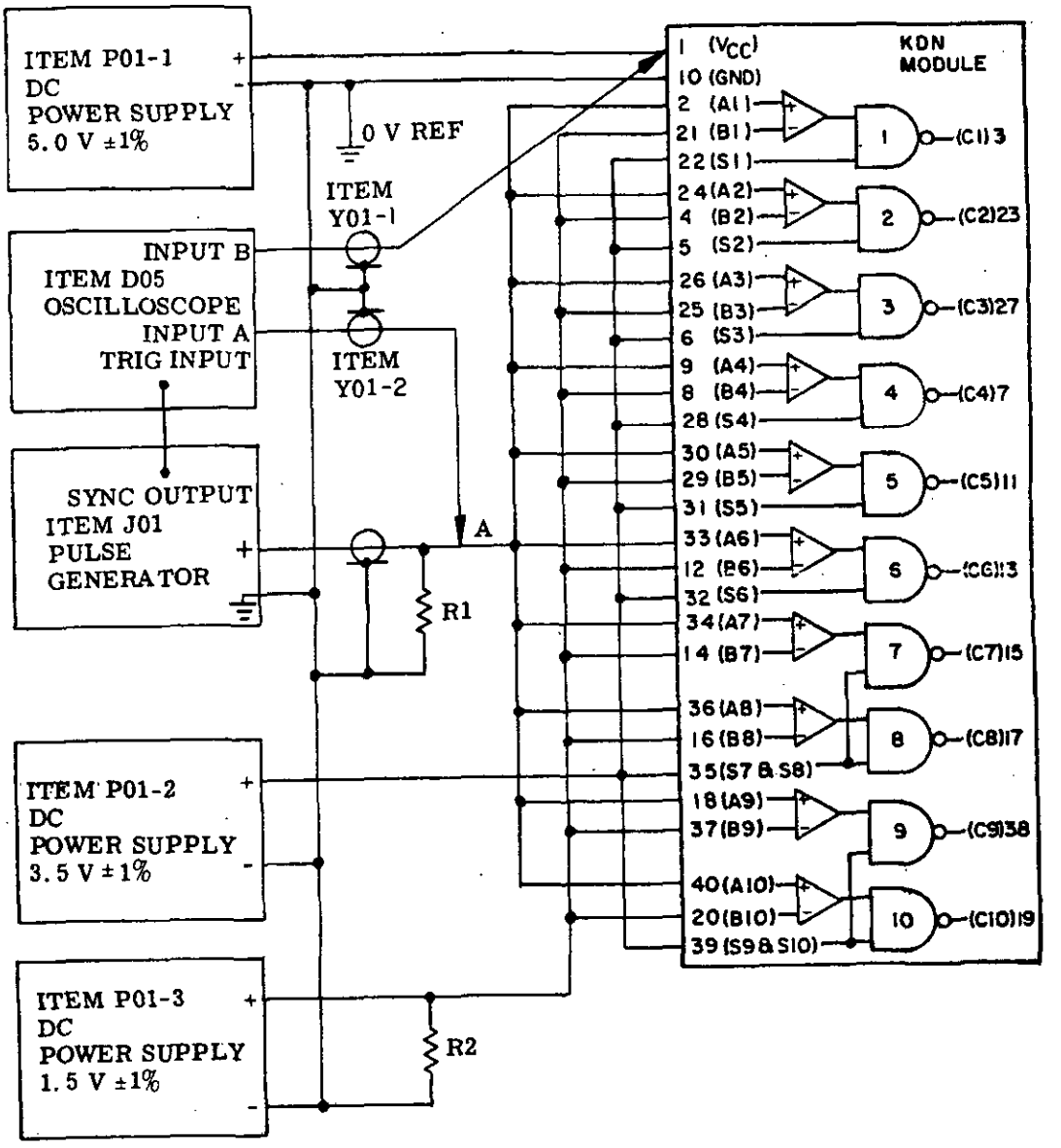
FIGURE 17. Output crosstalk test circuit waveforms for tables XVII and XVIII.

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TABLE XIX. Power supply transient amplitude test.

Test condition	Parameter	Initial limits 25° ±5°C Ambient		Unit	Test figure
		Min	Max		
X	Power supply transient amplitude	0	500	mV	18,19

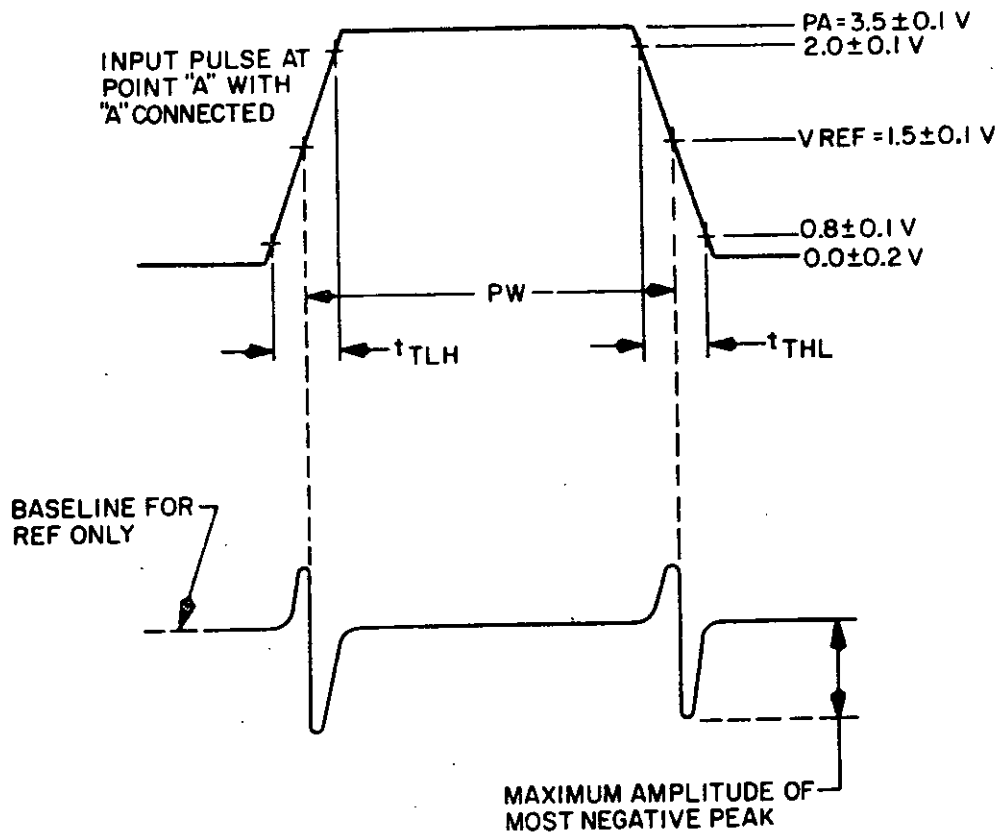
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NOTE: R1 = R2 = 51Ω ±5%, 1 W.

FIGURE 18. Power supply transient amplitude test circuit for table XIX.

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NOTES: Input pulse parameters:
1. PW = 500 ns ±10%.
2. $t_{TLH} = t_{THL} = 3.5 \pm 1.5$ ns (verified at each input pin).
3. PRR = 1 MHz ±10%.

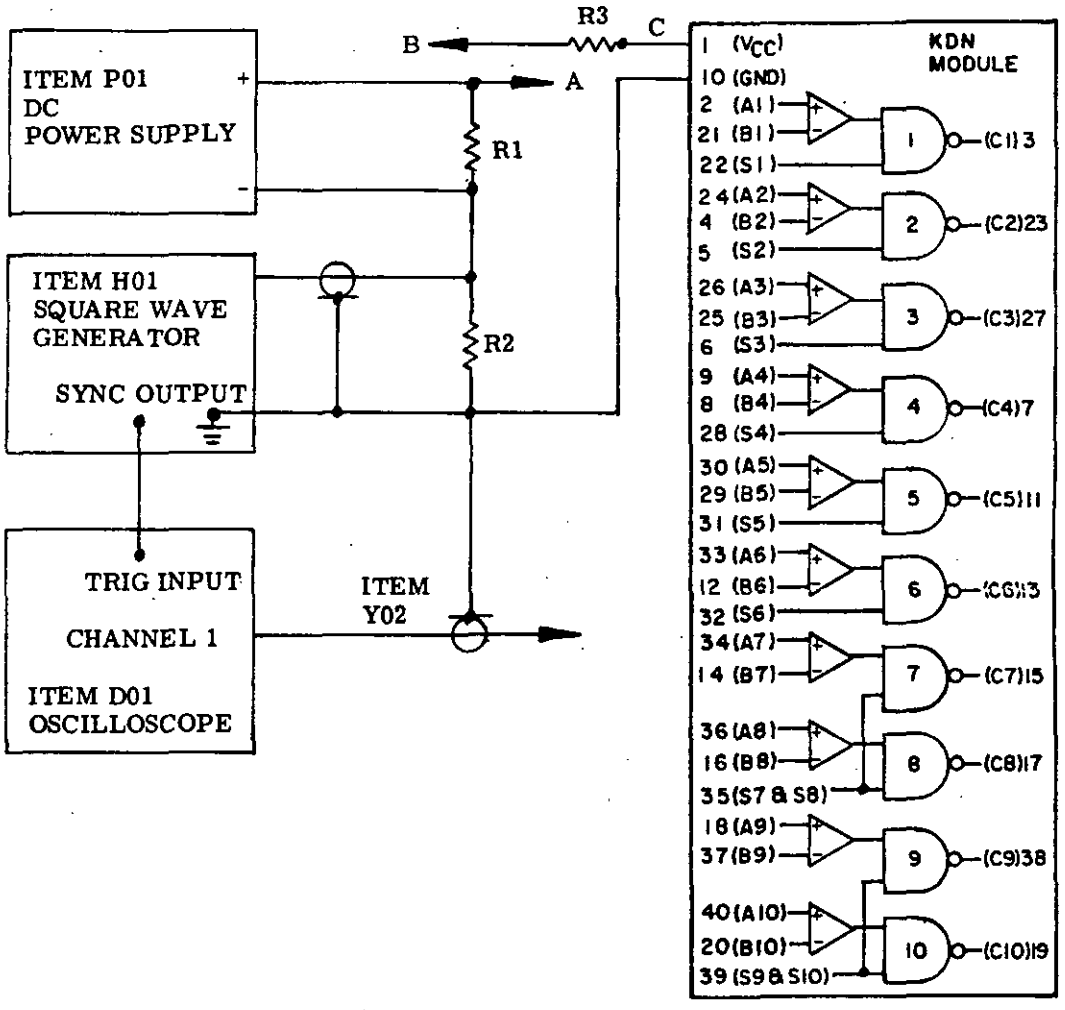
FIGURE 19. Power supply transient amplitude test circuit waveforms for table XIX.

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TABLE XX. Power supply transient rise time test.

Test condition	Parameter	Initial limits 25° ±5°C Ambient		Unit	Test figure
		Min	Max		
Y	Power supply transient rise time	1.0	2.0	ms	20,21

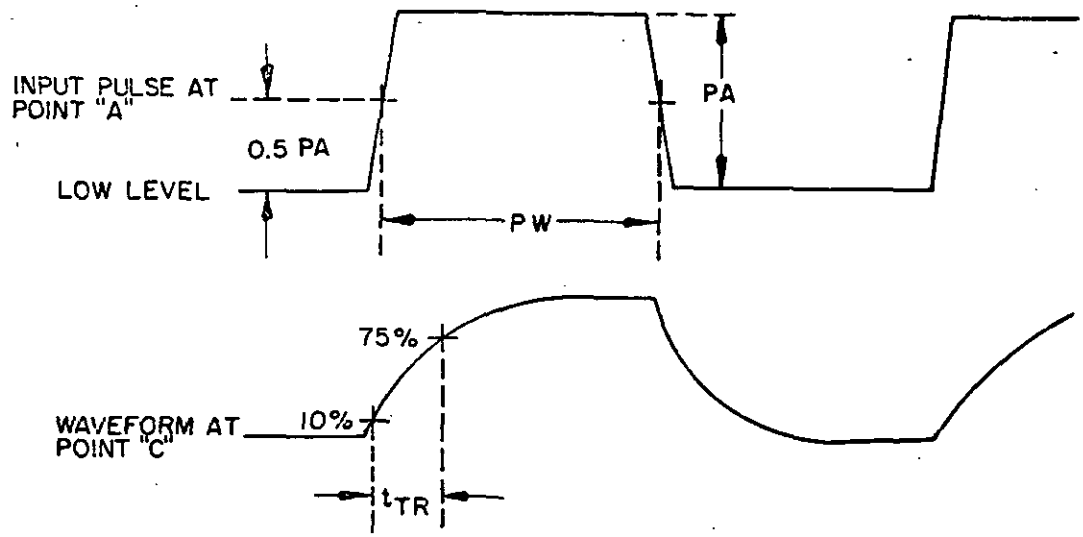
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NOTES:
1. R1 = R2 = 51Ω ±5%, 1 W.
2. R3 = 511Ω ±1%, 1/4 W.

FIGURE 20. Power supply transient rise time test circuit for table XX.

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NOTES:
1. PA = 0.5 V ±1% P-P.
2. PW = 10 ms ±10%.
3. PRR = 50 Hz (reference).
4. Rise and fall time of input $<$ 100 ns (reference).
5. With point "A" open, adjust input as shown.
6. With point "A" open, adjust power supply P01 such that the low level of the input is at 0 ±0.01 V.
7. Connect "A" to "B"; recheck the low level and pulse parameters at point "A".
8. Measure transient rise time (t_{TR}) at point "C".

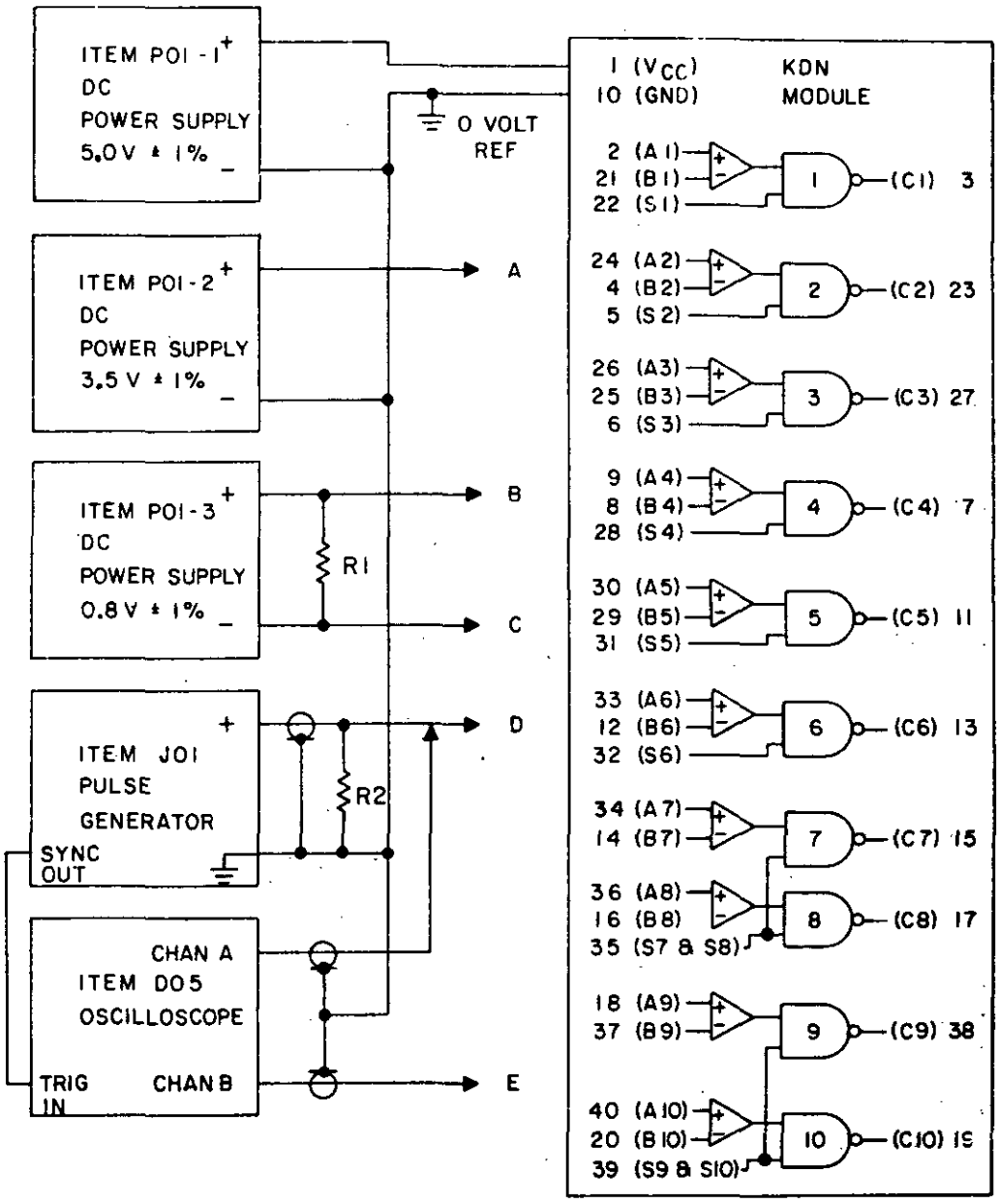
FIGURE 21. Power supply transient rise time test circuit waveforms for table XX.

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TABLE XXI. *Input pulse immunity (high-level and low-level) check test.*

Test condition	Parameter	Connect					Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
Z1	1A Input pulse immunity, high level, check	22	21	10	2	3	0.0	0.4	V	22, 23
Z2	2A	5	4		24	23				
Z3	3A	6	25		26	27				
Z4	4A	28	8		9	7				
Z5	5A	31	29		30	11				
Z6	6A	32	12		33	13				
Z7	7A	35	14		34	15				
Z8	8A	35	16		36	17				
Z9	9A	39	37		18	38				
Z10	10A Input pulse immunity, high level, check	39	20	10	40	19				
AA1	1A Input pulse immunity, low level, check	22	10	21	2	3				
AA2	2A	5		4	24	23				
AA3	3A	6		25	26	27				
AA4	4A	28		8	9	7				
AA5	5A	31		29	30	11				
AA6	6A	32		12	33	13				
AA7	7A	35		14	34	15				
AA8	8A	35		16	36	17				
AA9	9A	39		37	18	38				
AA10	10A Input pulse immunity, low level, check	39	10	20	40	19	0.0	0.4	V	22, 23

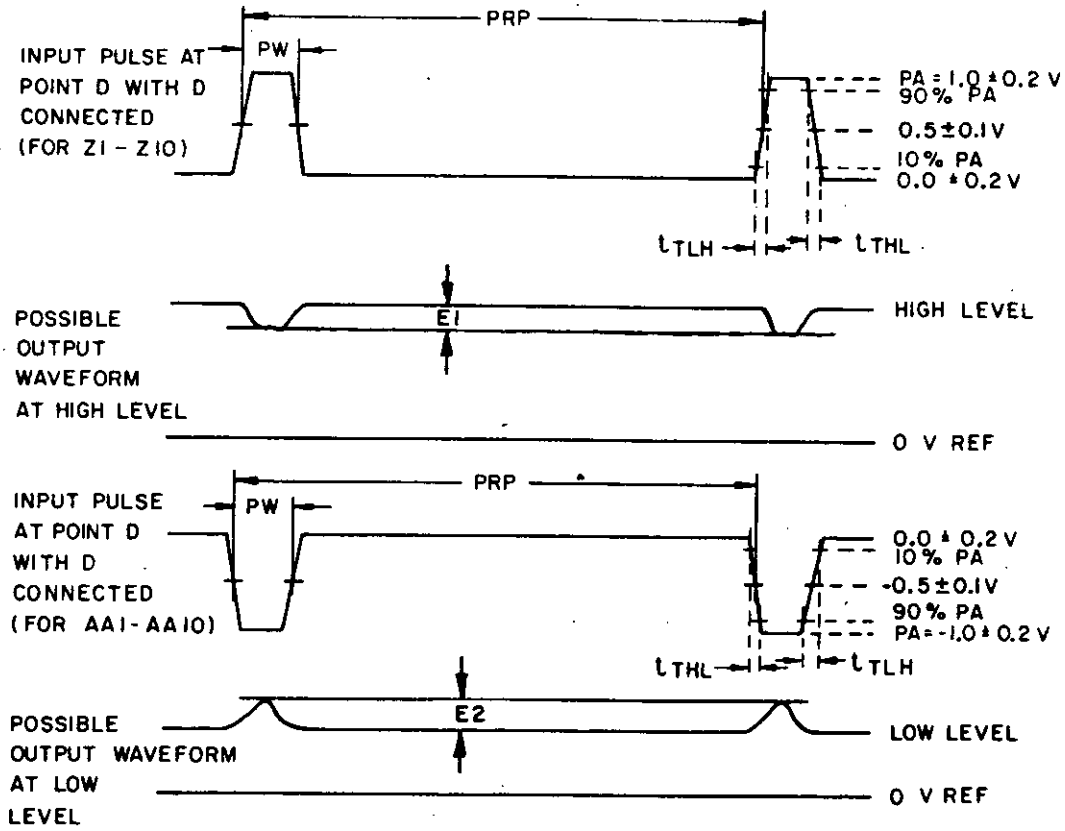
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NOTES:
1. R1 = R2 = 51Ω ±5%, 1 W.
2. Load output under test with load as specified on figure 10.

FIGURE 22. Input pulse immunity (high-level and low-level) check test circuit for table XXI.

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NOTES:
1. Input pulse parameters:
   PW = 10 ±1 ns.
   PRP = 200 ±20 ns.
   $t_{TLH} = t_{THL} = 4 \pm 1$ ns.
2. E1 is measured from the high level to the most negative peak of the high level waveform.
3. E2 is measured from the low level to the most positive peak of the low level waveform.

* FIGURE 23. Input pulse immunity (high-level and low-level) check test waveforms for table XXI.

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